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XIN Xiao-Ge, WU Tong-Wen, LI Jiang-Long, WANG Zai-Zhi, LI Wei-Ping, and WU Fang-Hua

Beijing Climate Center, China Meteorological Administration, Beijing 100081, China

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Abstract The historical simulation of phase five of the Coupled Model Intercomparison Project (CMIP5) experiments performed by the Beijing Climate Center climate system model (BCC_CSM1.1) is evaluated regarding the time evolutions of the global and China mean surface air temperature (SAT) and surface climate change over China in recent decades. BCC_CSM1.1 has better capability at reproducing the time evolutions of the global and China mean SAT than BCC_CSM1.0. By the year 2005, the BCC_CSM1.1 model simulates a warming amplitude of approximately 1°C in China over the 1961–1990 mean, which is consistent with observation. The distributions of the warming trend over China in the four seasons during 1958–2004 are basically reproduced by BCC_CSM1.1, with the warmest occurring in winter. Although the cooling signal of Southwest China in spring is partly reproduced by BCC_CSM1.1, the cooling trend over central eastern China in summer is omitted by the model. For the precipitation change, BCC_CSM1.1 has good performance in spring, with drought in Southeast China. After removing the linear trend, the interannual correlation map between the model and the observation shows that the model has better capability at reproducing the summer SAT over China and spring precipitation over Southeast China.

Keywords: BCC_CSM, climate system model, simulation, climate change

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1 Introduction

Observational analyses have indicated that distinctive climate changes have occurred over China in the recent half century, including spring cooling in southwestern China, summer cooling in central eastern China, spring drought in southeastern China, and a "northern drought and southern flood" (NDSF) pattern in summer (e.g., Yu et al., 2004; Li et al., 2005; Xin et al., 2006, 2008; Yu and Zhou, 2007; Zhou et al., 2009a). Many studies have investigated the causes of climate changes in China. These changes include warming in the western Pacific Ocean and the Indian Ocean (Gong and Ho, 2002; Zhou et al., 2009b), warming of the whole tropical ocean (Zhou et al., 2008; Li et al., 2010), teleconnection related to the winter North Atlantic Oscillation (Xin et al., 2006, 2010; Li et al., 2008), and aerosols such as black carbon (Menon et al., 2002). This reflects the complex processes related to the long-term variations of climate in China, as different factors may interact with each other. Climate system models are an important tool for identifying the internal relationships between different factors. However, the reproduction of the climate change in recent decades over East Asia is a challenge when using climate models participiated in phase three of the Coupled Model Intercomparison Project (CMIP3). Few models can realistically simulate the cooling trend and NDSF rainfall change in East China (Zhou and Yu, 2006; Sun and Ding, 2008).

After phase 3 of CMIP, modeling groups worldwide invested great efforts in improving climate system models, including the physical parameterization of the physical processes, their resolution, and for more complex processes. The Beijing Climate Center (BCC) has also devoted much effort to developing a new generation coupled climate system model (BCC_CSM1.1). This model was used in the recent phase 5 of CMIP (CMIP5) and has performed most prescribed experiments (Taylor et al., 2011). However, the model's performance in capturing climate change in China has never been evaluated. The BCC_CSM1.1 model developed by the BCC will be frequently used in climate prediction and projection in China. The performance of this model at reproducing climate variations in China is the basis for credible geographical distributions of future climate predictions and projections. The main motivation of this paper is to examine the extent to which the BCC_CSM1.1 model can reproduce the observed climate variations of surface air temperature (SAT) and precipitation in China in recent decades. Time evolutions of the global and China mean SAT in the 20th century are also evaluated to provide a basic reference for the simulation ability of this model.

The other parts of this paper are organized as follows. The model description and experimental design are described in section 2. Section 3 presents the results simulated by BCC_CSM1.1. Conclusions are given in section 4.

2 Model description and experimental design

The BCC_CSM1.1 is a coupled climate system model including atmosphere, ocean, land surface, and sea ice components, which are coupled through the National Center for Atmospheric Research (NCAR) coupler version 5. The atmospheric component, BCC_AGCM2.1, is

Corresponding author: XIN Xiao-Ge, xinxg@cma.gov.cn

a spectral model with a horizontal T42 truncation (~2.8°) and 26 layers in the vertical. The model is described in detail by Wu et al. (2008, 2010). The land surface model, BCC_AVIM1.0 (Atmosphere and Vegetation Interaction Model version 1), is originated from AVIM2 (Ji et al., 2008). The ocean model, Modular Ocean Model version 4 (MOM4_L40), was developed by the Geophysical Fluid Dynamics Laboratory (GFDL). The horizontal resolution is (1/3)°×1° with a tripolar grid and a vertical resolution of 40 levels. The sea ice component of BCC_CSM1.1 is the Sea Ice Simulator (SIS) from GFDL, which has the same horizontal resolution as MOM4_L40. This model incorporates an interactive global carbon cycle process fulfilling the requirements of CMIP5.

The BCC_CSM1.1 model was run for a pre-industrial control (piControl) simulation for 500 years with the atmospheric forcing fixed at the year 1850. Three ensemble members of a historical simulation from 1850 to 2005 were performed with the initiating states from different points of piControl. In this paper, the ensemble mean of the three members is used in the analysis. In the historical simulation, the external forcing changes with time and includes well-mixed greenhouse gases, ozone, aerosols, solar radiation, and volcanoes. Those forcing data, except volcanoes, are all derived from the CMIP5 website, thus providing common external forcing for the climate models in the CMIP5 experiments. The volcanic dataset is consistent with the NCAR Community Climate System Model version 3 (CCSM3) in the 20th century simulation for CMIP3, which is included in the model as aerosol optical depths (Ammann et al., 2003; Meehl et al., 2006). The greenhouse gases cover CO_2, N_2O, CH_4, CFC_{11}, and CFC_{12}. The aerosol properties include sulfate from natural and anthropogenic sources, sea salt, black and organic carbon, and soil dust.

For comparison, the 20th century simulation (20C3M) during 1870–1999 by the earlier version of the model, BCC_CSM1.0, is used. The main difference between BCC_CSM1.1 and BCC_CSM1.0 is that the latter uses different land surface (Community Land Surface Model version 3, CLM3), ocean (Parallel Ocean Program, POP) and sea ice (Community Sea Ice Model version 4, CSIM4) models. A description of BCC_CSM1.0 and the forcing data used in 20C3M can be found in Zhang et al. (2011).

The global mean time series of SAT during 1850–2005 is from the HadCRUT3 dataset. The dataset is a collaborative product of the Met Office Hadley Centre and the Climatic Research Unit at the University of East Anglia (Brohan et al., 2006). The time series of mean SAT over China during 1906–2005 is from Tang and Ren (2005). A station observation dataset of 560 sites in China from 1958 to 2004 is used in the evaluation of the distribution of surface climate change in China and was used in Xin et al. (2010). The Chinese-domain-averaged SAT and precipitation for the model results is defined as the area-weighted average of three rectangular boxes: (28–50°N, 80–97.5°E), (22.5–43°N, 97.5–122.5°E), and (43–54°N, 117.5–130°E), as used in Zhou and Yu (2006).

3 Results

3.1 Time evolutions of global and China mean SAT

Earlier studies showed that climate models are able to reproduce the time evolution of observed global and China averaged SAT with the external forcing of natural and anthropogenic factors (Zhou and Yu, 2006). The simulation of BCC_CSM1.1 and BCC_CSM1.0 is evaluated and presented in Fig. 1. Time evolutions of global mean SAT in both models are basically consistent with the observation, but the BCC_CSM1.1 results are closer to the observation than those of BCC_CSM1.0. During the common period (1870–1999) of the two models, the correlation coefficients with the observation are 0.86 and 0.70 for BCC_CSM1.1 and BCC_CSM1.0, respectively. It is noted that the warming trend simulated by BCC_CSM1.0 in the latter half of the 20th century is higher than BCC_CSM1.1.

The linear trend of the global mean SAT simulated by BCC_CSM1.1 during 1850–2005 is 0.66°C/100 yr, which is larger than the observed value of 0.42°C/100 yr. By 2005, the simulated warming amplitude in BCC_CSM1.1 relative to the 1961–1990 mean is 0.7°C, which is 0.2°C higher than the observation. The overestimation of the

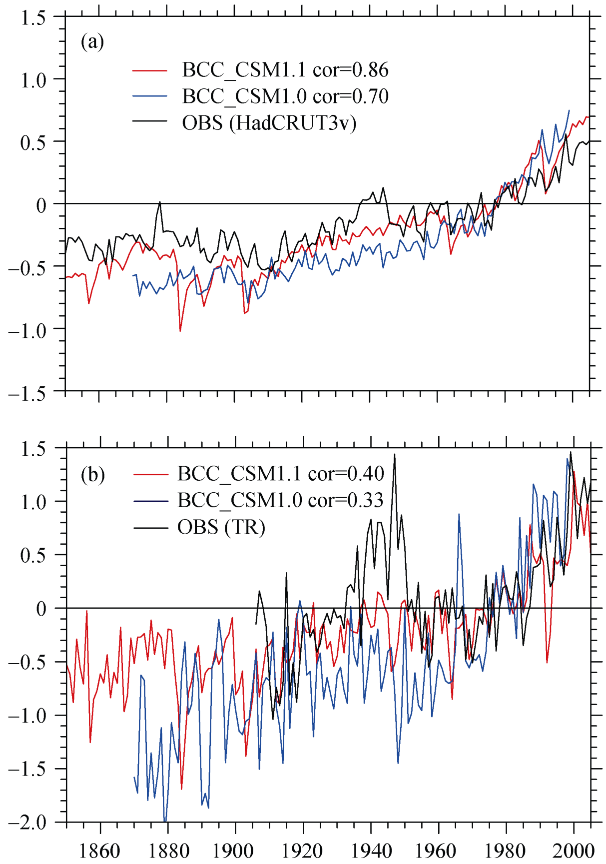

Figure 1 Anomalies of mean surface air temperature averaged over (a) the globe and (b) China relative to the 1961–1990 mean simulated by BCC_CSM1.1 and BCC_CSM1.0. The global observation data are from HadCRUT3. The China observation data are from Tang and Ren (2005). Units: °C.

trend by the model mainly occurred in the late 20th century. During 1966–2005, the linear trend of the BCC_CSM1.1 is 2.39°C/100 yr, while the observation is 1.61°C/100 yr. BCC_CSM1.1 could not reproduce the warming peak in the mid-1940s, which is a common bias in earlier simulations of climate models (Zhou and Yu, 2006). Meehl et al. (2006) considered that the model's deficiency in simulating this warming peak is related to the uncertainty regarding the solar forcing data reconstructions.

The China mean SAT varies consistently with the global feature, except that the variability in China is greater (Fig. 1). During the common simulation period of 1906–1999, the evolution of the China mean SAT simulated by BCC_CSM1.1 (0.40) is more closely correlated with the observation than that by BCC_CSM1.0 (0.33). By the year 2005, both the observation and BCC_CSM1.1 show a warming amplitude at approximately 1°C. The warming peak in the mid-1940s in China does not appear in the model. During the two warming periods of 1905–1941 and 1966–2005, the observed linear trend of the China SAT is 2.66°C/100 yr and 3.7°C/100 yr. However, the BCC_CSM1.1 model underestimated the warming trends of both periods, with values of 1.44°C/100 yr and 2.66°C/100 yr, respectively. The underestimation of the warming trend during 1905–1941 by BCC_CSM1.1 is attributed to the model being unable to reproduce the warming peak in the mid-1940s. The lower warming during 1966–2005 in BCC_CSM1.1 in China is opposite to that of the global mean, the reasons for which will be explored in further studies using attribution experiments.

3.2 Climate change in China

Figure 2 shows the linear trend of SAT during 1958–

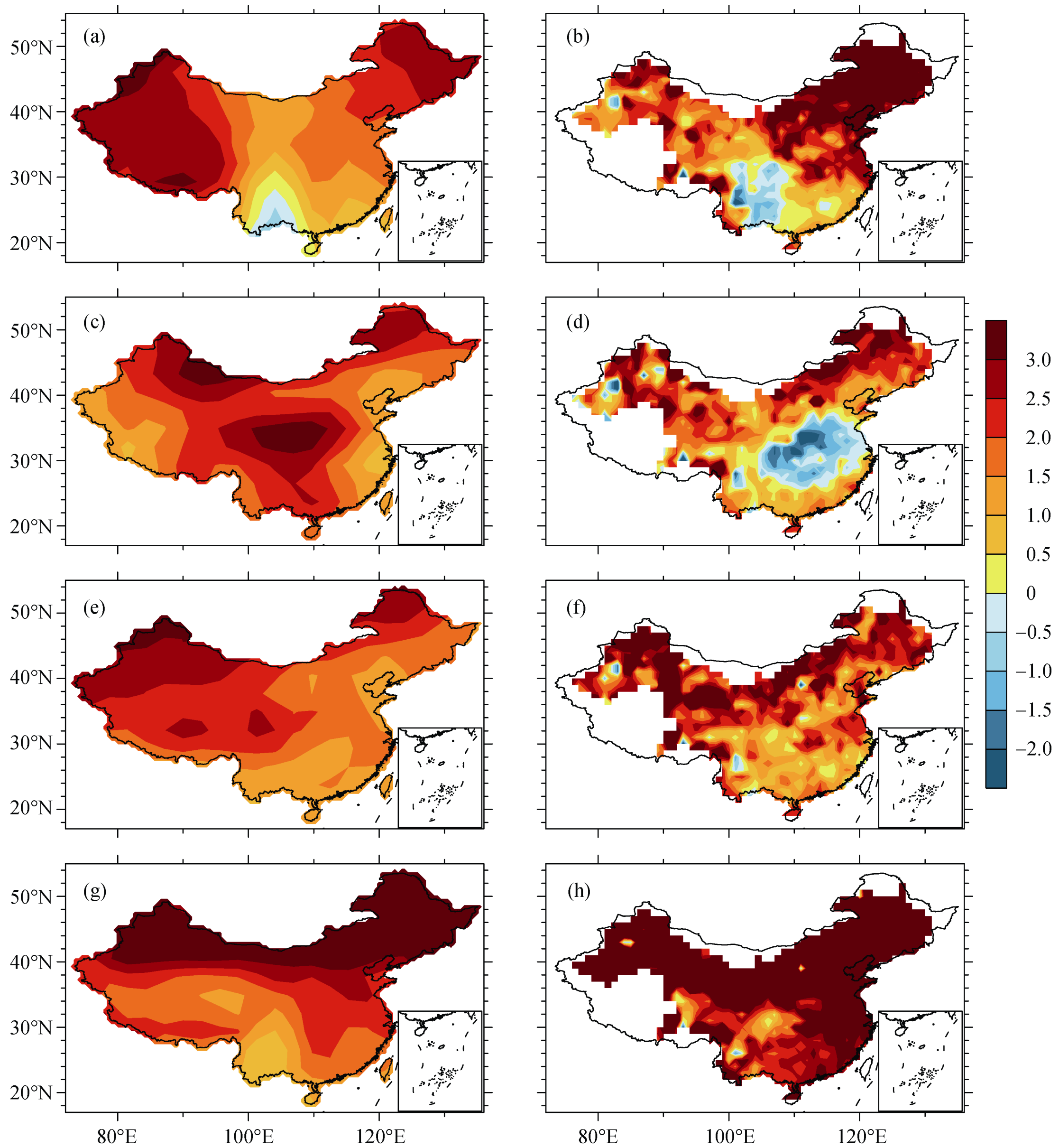

Figure 2 Linear trend of seasonal mean surface air temperature during 1958–2004 in (a, b) spring, (c, d) summer, (e, f) autumn, and (g, h) winter for BCC-CSM1.1 (left column) and station observation (right column). Units: °C/100 yr.

2004 in the four seasons simulated by BCC_CSM1.1 in comparison with the station observation in China. The model can generally simulate the warming trend in China. In both the simulation and observation, winter is the warmest season, with the linear trend over most parts of China being greater than 3°C/100 yr. The distributions of the warming trend in autumn and winter are close to the observation, except that the model underestimates the trend in some regions. In spring and summer, the observation shows regional cooling signals, including the spring cooling in southwestern China and summer cooling in central east China. BCC_CSM1.1 produces a cooling signal in spring, which is to the south of that in the observation. The summer cooling in central East China is omitted by the model. Further examination shows that BCC_CSM1.1 cannot reproduce the upper tropospheric cooling in East Asia and the cooling trend in the North Pacific Ocean (figures not shown), which were suggested to be connected with the summer climate change in China (Yu et al., 2004; Yang et al., 2005; Yu and Zhou, 2007). It is inferred that the mechanism responsible for the cooling trend over central east China is not realistically simulated by BCC_CSM1.1. The physical processes in BCC_CSM1.1 need to be improved in the future.

In recent decades, the linear trend of precipitation in China has shown distinct features in each season. As shown in Fig. 3, the observations show drought in southeastern China in spring, which can be approximately simulated by BCC_CSM1.1, though with a weaker strength. However, the noticeable NDSF precipitation change in China could not be reproduced by the model. The observed precipitation change in autumn and winter in BCC_CSM1.1 could not be generated by the model (figure omitted).

Overall, the BCC_CSM1.1 model performs well in reproducing the temperature change in recent decades, except for the summer cooling in central east China. For the precipitation change, the model only shows reasonable simulation in the spring drought over southeastern China.

3.3 Interannual variations of surface climate in China

After removing the linear trend in both BCC_CSM1.1 and the observation, the correlation between them reflects whether the interannual variability could be reproduced by the model. Figure 4 shows the correlation map of SAT and precipitation between BCC_CSM1.1 and the observation in spring and summer during 1958–2004 after linearly detrending both datasets. As can be seen, the SAT in summer over a large area of China is significantly above the 5% level, including Northeast China, western China, parts of North China, and southeastern China. In spring, only part of the Huanghuai valley is above the 5% significant level. For precipitation, spring is the season with the largest significant area, including the western part of the Tibetan Plateau, and central and southeastern China. In summer, a significant correlation appears in part of southeastern China. In consideration of both the linear trend and interannual variations, the projection of spring precipitation over Southeast China by BCC_CSM1.1 is most credible for precipitation among all seasons.

4 Conclusions

The climate system model BCC_CSM1.1 is evaluated in the simulation of climate variations over East China. The time evolution of the global and China mean SAT is better simulated by BCC_CSM1.1 than BCC_CSM1.0.

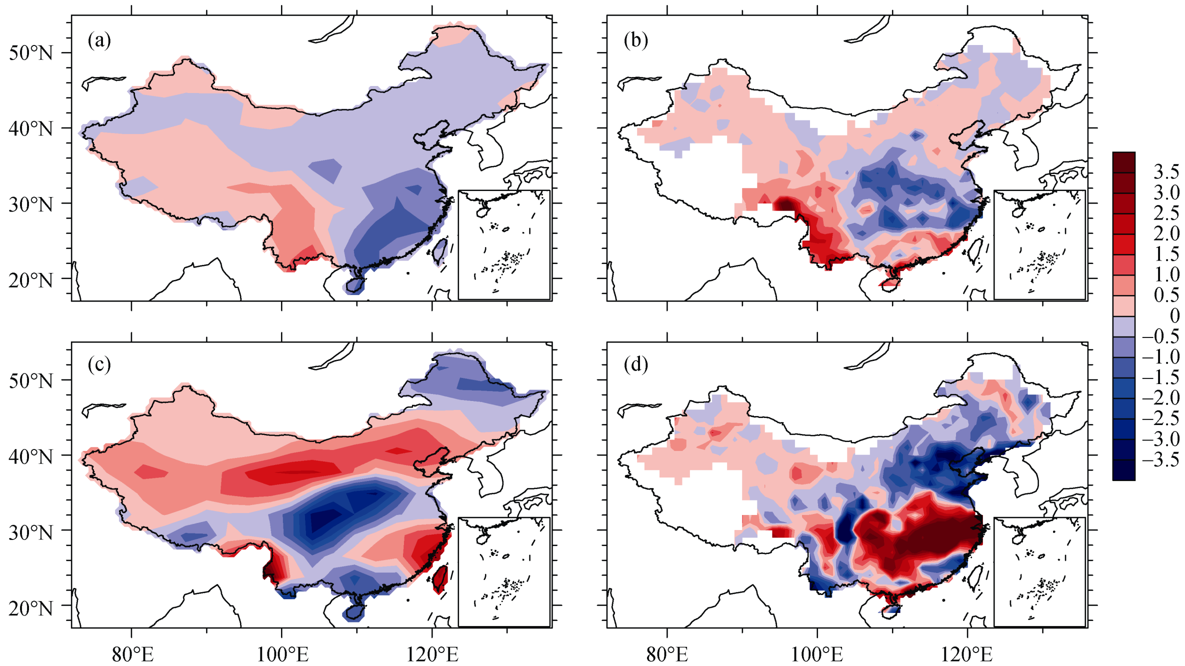

Figure 3 Linear trend of precipitation during 1958–2004 in (a, b) spring (MAM) and (c, d) summer (JJA) for BCC-CSM1.1 (left column) and station observation (right column). Units: mm d^{-1}/100 yr.

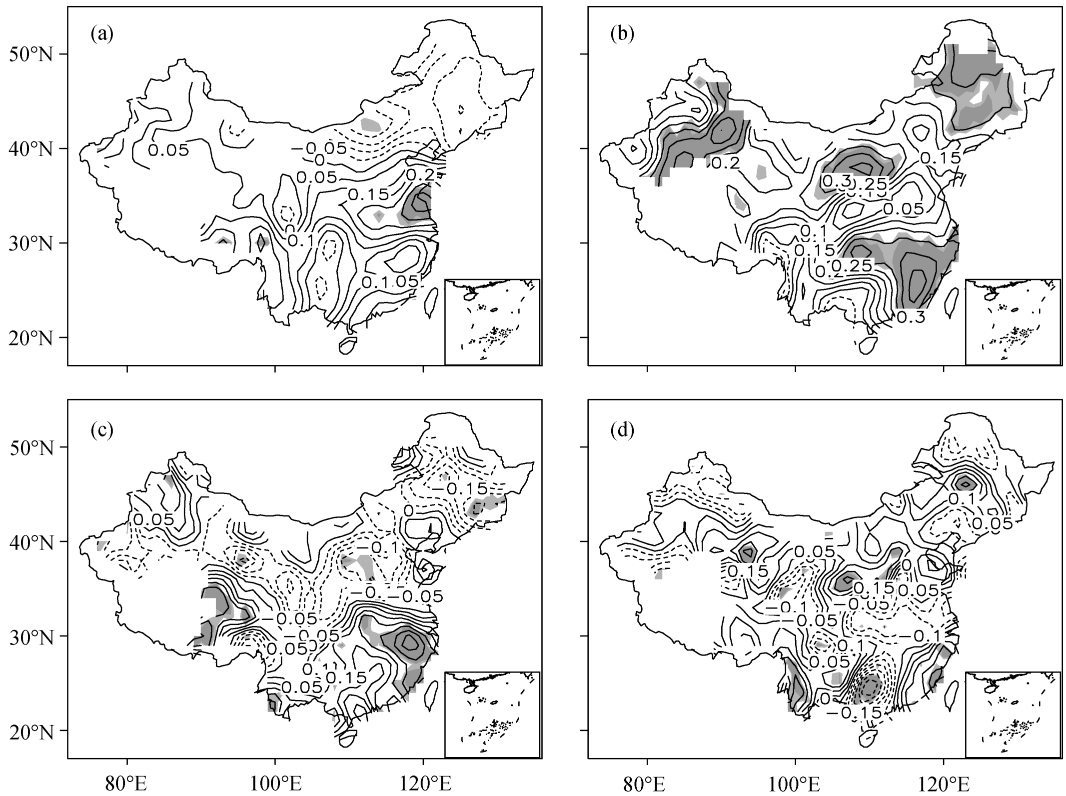

Figure 4 Correlation map between BCC_CSM1.1 and station observation for (a) spring and (b) summer surface air temperature and for (c) spring and (d) summer precipitation after linearly detrending during 1958–2004. The deep (light) shade indicates the correlation is significant at the 5% (10%) level.

By the year 2005, both the observation and BCC_CSM1.1 show a warming amplitude of approximately 1°C over the 1961–1990 mean in China. For recent decades, the distribution of the warming trend over China in each season is basically reproduced by BCC_CSM1.1. In both the simulation and observation, winter is the warmest season, with a linear trend over most parts of China greater than 3°C/100 yr. The spring cooling signal in southwestern China is partly reproduced by the model, but the summer cooling in central east China could not be generated. The model has better capability at reproducing the drought over southeast China in spring. The summer precipitation change with the NDSF pattern in China could not be reproduced by the model. The model could realistically simulate the interannual variability of the summer SAT in most parts of China and the spring precipitation over Southeast China. In consideration of both the linear trend and interannual variations of the precipitation, the projection of spring precipitation over Southeast China by BCC_CSM1.1 is the most credible among all seasons.

The regional features in BCC-CSM1.1 need to be improved in the future. Improvement of the horizontal resolution is one way for the model to recognize regional climate features. The BCC is developing a higher-resolution model, with atmospheric model horizontal resolutions of T106 and T266. The T106 resolution model BCC_CSM1.1_M is being used to perform core experiments of CMIP5. It is worth examining whether the higher-resolution model has better capability at simulating China's climate variations.

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